



Using MIT's new system, supervisors can correct a robot's mistakes by simply thinking about it (Image courtesy: MIT)

the rooms at the university's Robotics, Perception and Learning Lab, and maps these in detail. It uses a depth camera (RGB-D) to grab points of physical space and dump these into a database, from which 3D models of the rooms can be generated.

According to a news report, "The

system KTH researchers use detects objects to learn by modelling the static part of the environment and extracting dynamic elements. It then creates and executes a view plan around a dynamic element to gather additional views for learning. This autonomous learning process enables Rosie to distinguish dynamic elements from static ones and perceive depth and distance." This helps the robot understand where things are and negotiate physical spaces.

Just a thought can bring the robot back on track

It is one thing for robots to learn to work autonomously, it is another for them to be capable of working with humans. Some consider the latter to be more difficult. To be able to co-exist, robots must be able to move around safely with humans (as in the case of KTH's Rosie) and also understand what humans want, even when the instruction or plan is not clearly, digitally explained to the robot.

Explaining things in natural language is never foolproof because each person has a different way of communicating. But if only robots could understand what we think, the problem would be entirely solved. As a step towards this, Massachusetts Institute of Technology's Computer Science and Artificial Intelligence Laboratory (CSAIL) and Boston University are creating a feedback system that lets you

correct a robot's mistakes instantly by simply thinking about it.

The experiment basically involves a humanoid robot called Baxter performing an object-sorting task and a human watching it. The person watching the robot has to wear a special head gear. The system uses an electroencephalography monitor to record the person's brain activity. A novel machine learning algorithm is applied to this data to classify brain waves in the space of 10 to 30 milliseconds. When the robot indicates its choice, the system helps it to find out whether the human agrees with the choice or notices an error. The person watching the robot does not have to gesture, nod or even blink. He or she simply needs to agree or disagree mentally with the robot's action. This is much more natural than earlier methods of controlling robots with thoughts.

The team lead by CSAIL director Daniela Rus has managed to achieve this by focusing the system on brain signals called error-related potentials (ErrPs), which are generated whenever our brains notice a mistake. When the robot indicates the choice it is about to make, the system uses ErrPs to understand whether the human supervisor agrees with the decision.

According to the news report, "ErrP signals are extremely faint, which means that the system has to be fine-tuned enough to both classify the signal and incorporate it into the feedback loop for the human operator."

Additionally, the team has also worked on the possibility of the system not noticing the human's original correction, which might lead to secondary errors. In such a case, if the robot is not sure about its decision, it can trigger a human response to get a more accurate answer. Further, since ErrP signals appear to be proportional to how bad the mistake is, future systems could be extended to work for more complex multi-choice tasks.

Matić is building such robots through her startup Embodied Inc. Research and trials have shown promising results. One study found that autistic children showed more autonomous behaviour upon copying the motions of socially-assistive robots.

In another study, patients recovering from stroke responded more quickly to upper-extremity exercises when prompted and motivated by socially-assistive robots.

Robots can mingle in crowded places, too

Movement of robots was once considered a mechanical challenge. Now, scientists have realised it has more to do with intelligence. For a robot to move comfortably in a crowded place like a school or an office, it needs to first learn things that we take for granted. It needs to learn the things that populate the space, which of these things are stationary and which ones move, understand that some of these things move only occasionally while others move frequently and suddenly, and so on. In short, it needs to autonomously learn its way around a dynamic environment. This is what a team at KTH Royal Institute of Technology in Stockholm hopes to achieve.

Rosie (yes, we know it sounds familiar) is a robot in their lab that has already learnt to perceive 3D environments, move about and interact safely in these. Rosie repeatedly visits